Final Exam

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1. Consider the IBVP:

$$u_{tt} - u_{xx} = 0,\ u(0,t) = 0,\ u(x,0) = 0, u_t(x,0) = f(x), \tag{1}$$

where $x > 0$, $t > 0$, $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 0$ for $x > 1$.

Proof (Hernandez). The solution is given by d'Alembert's formula below:

$$\begin{aligned} u(x,t) &= \frac{1}{2}\left[0+0\right] + \frac{1}{2\cdot 1}\int_{x-t}^{x+t} f(\xi)\ d\xi \\ &= 0 + \frac{1}{2}\int_{\min(x-t,0)}^{\min(x+t,1)} f(\xi)\ d\xi. \end{aligned}$$

□

2.

$$u_t - u_{xx} = \frac{1}{x^2+1},\ \ -\infty < x < \infty, t > 0, u(x,0) = 0. \tag{2}$$

Proof (Hernandez). According to Duhamel's principle, the solution is given below:

$$u(x,t) = \int_0^t \int_{-\infty}^{\infty} \frac{G(x-y, t-\tau)}{y^2+1}\ dyd\tau$$

□

3.

$$u_{tt} - u_{xx} = x,\ u(x,0) = 0,\ u_t(x,0) = 1, u_x(0,t) = u_x(\pi,t) = 0, \tag{3}$$

where $0 < x < \pi$, $t > 0$.

Proof (Hernandez). The solution is given by a superposition $u = v + w$ solving:

$$\begin{aligned} v_{tt} - v_{xx} = 0,\ v(x,0) = 0,\ v_t(x,0) = 1, v_x(0,t) = v_x(\pi,t) = 0, \\ w_{tt} - w_{xx} = x,\ w(x,0) = 0,\ w_t(x,0) = 0, w_x(0,t) = w_x(\pi,t) = 0, \end{aligned}$$

Clearly, $v(x,t) = t$. For w, we obtain the following eigenfunctions:

$$\begin{aligned} -\varphi_n''(x) &= \lambda^2\varphi_n(x), & \varphi_(0) &= \varphi(\pi) = 0, \\ \varphi_n(x) &= \sin(\lambda x), & \lambda &= \frac{n\pi\cdot 1}{\pi} = n, \end{aligned}$$

which leads to the following solution:

$$w(x,t) = \sum_{n=1}^{\infty}\left(\frac{1}{n}\int_0^t f_n(s)\cos(ns)\ ds\right)\sin(nt)\sin(nx),$$

where $f_n(s)$ is the Fourier coefficient of x as shown below:

$$x = \sum_{n=1}^{\infty} f_n(t)\sin(nx).$$

□

4.

$$u_{tt} + u_{xxxx} = 0 \tag{4}$$

Proof (Hernandez). We apply separation of variables. □

5.

$$u_{tt} = u_{xx} - 2u,\ u_x(0,t) = 0, u(1,t) = 0, u(x,0) = 1, u_t(x,0) = 0, \tag{5}$$

where $0 < x < 1$, $t > 0$.

Proof (Hernandez). The solution is given by the superposition $u = v + w$ solving:

$$v_{tt} - v_{xx} = -2v,\ v_x(0,t) = 0, v(1,t) = 0, v(x,0) = 0, v_t(x,0) = 0,$$
$$w_{tt} - w_{xx} = 0,\ w_x(0,t) = 0, w(1,t) = 0, w(x,0) = 1, w_t(x,0) = 0,$$

where $v = 0$ everywhere and w is given by d'Alembert's formula below:

$$w(x,0) = \frac{1}{2}\left[1 + 1\right] + \frac{1}{2 \cdot 1}\int_{x-ct}^{x+ct} 0 \; d\xi = 1.$$

□

6.

$$\frac{1}{2x}u_x + u_t - u = 0,\ \ -\infty < x < \infty, t > 0, u(x,0) = x^2 \tag{6}$$

Proof (Hernandez). We obtain the following solution:

$$u(x,t) = -\frac{t - x^2}{e^t}.$$

□

7.

$$(x+t)u_x + tu_t + u = 0,\ \ -\infty < x < \infty,\ t > 1,\ u(x,1) = x. \tag{7}$$

Proof (Hernandez). We apply the method of characteristics. □

8. Consider the following IBVP in the unit 3-d sphere:

$$u_{tt} = \Delta u, 0 < r < 1, u|_{r=1} = 0,\ u|_{t=0} = r, u_t|_{t=0} = 1. \tag{8}$$

Proof (Hernandez). Our equation is $u_{tt} = \Delta u = u_{rr} + \frac{2}{r}u_r$, giving a solution of the following form:

$$u(r,t) = \sum_{n=1}^{\infty} c_n(t) y_n(r), \qquad y_n(r) = \frac{\sin(n\pi r)}{r},$$

and eventually obtain:

$$u(r,t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2T_0}{n} e^{-n^2 kt} \frac{\sin(nr)}{r}.$$

□